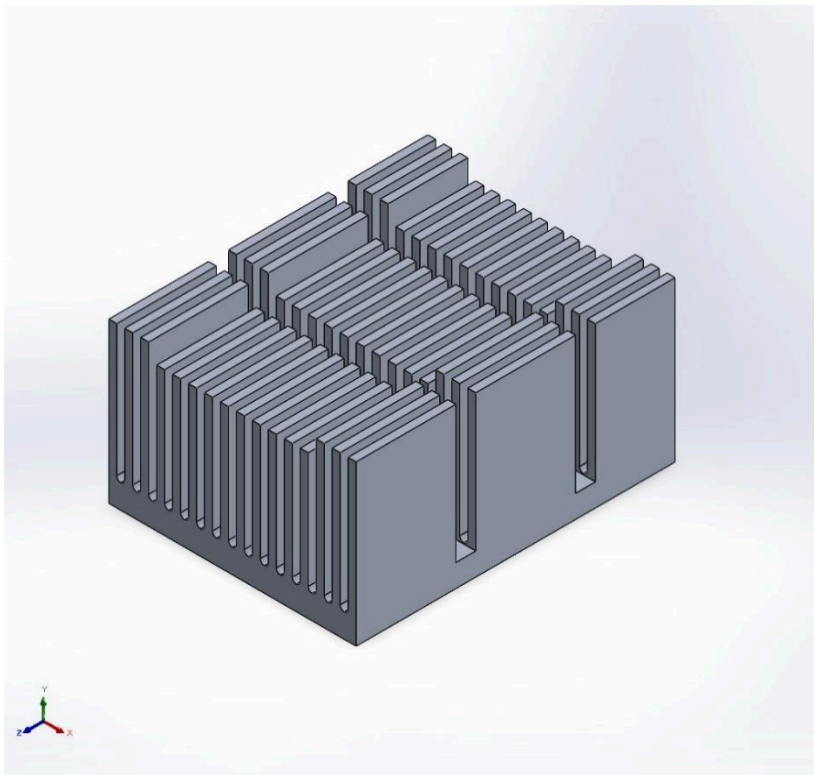


Heatsink



Description

Steady-state thermal finite element analysis of an extruded finned aluminum heatsink subjected to uniform heat generation from an attached CPU. The study evaluates multi-directional conductive heat spreading through the baseplate, convective heat dissipation across the fin channels, steady-state temperature gradients, and localized heat flux concentrations under constant operating conditions.

Simulation of Heatsink

Date: Wednesday, December 10, 2025
Designer: LUIS RAMIREZ
Study name: HEATSINK STEADY STATE ANALYSIS
Analysis type: Thermal(Steady state)

Table of Contents

Description	1
Assumptions	2
Model Information	3
Study Properties	5
Units	5
Material Properties	6
Thermal Loads	6
Interaction Information	7
Mesh information	7
Sensor Details	10
Study Results	10
Conclusion	15



Heatsink

Assumptions

- **Fourier Conduction and Isotropic Material:** Heat transfer within the solid heatsink follows Fourier's Law of Conduction: $q_x = -k * (dT/dx)$. The 1060 Aluminum Alloy is assumed to be homogeneous and isotropic, with a constant thermal conductivity of $k = 200 \text{ W/(m-K)}$ across all directions.
- **Steady-State Energy Balance:** The system is in steady-state thermal balance ($dT/dt = 0$), meaning total heat power in equals total convective heat out ($Q_{in} = Q_{out} = 50 \text{ W}$). Internal thermal energy storage is zero.
- **Uniform Heat Power Input:** The 50 W heat source is distributed evenly across the center split square face of area $A = 0.012 \text{ m} \times 0.012 \text{ m} = 0.000144 \text{ m}^2$.
- **Newton's Law of Cooling:** Heat rejection from all 239 exposed fin and body faces to the surrounding air follows Newton's Law of Cooling: $q = h * (T_s - T_{amb})$. A constant convection coefficient of $h = 25 \text{ W/(m}^2\text{-K)}$ and an ambient room temperature of $T_{amb} = 300 \text{ K}$ (27 C) are applied.
- **Zero Contact Resistance:** Ideal thermal contact is assumed between the CPU and the bottom mounting face (thermal resistance $R = 0$), modeling direct solid contact without thermal interface material resistance.
- **Negligible Radiation:** Surface-to-surface and surface-to-ambient radiation heat transfer are neglected because convective cooling dominates at these operating temperatures.



Heatsink

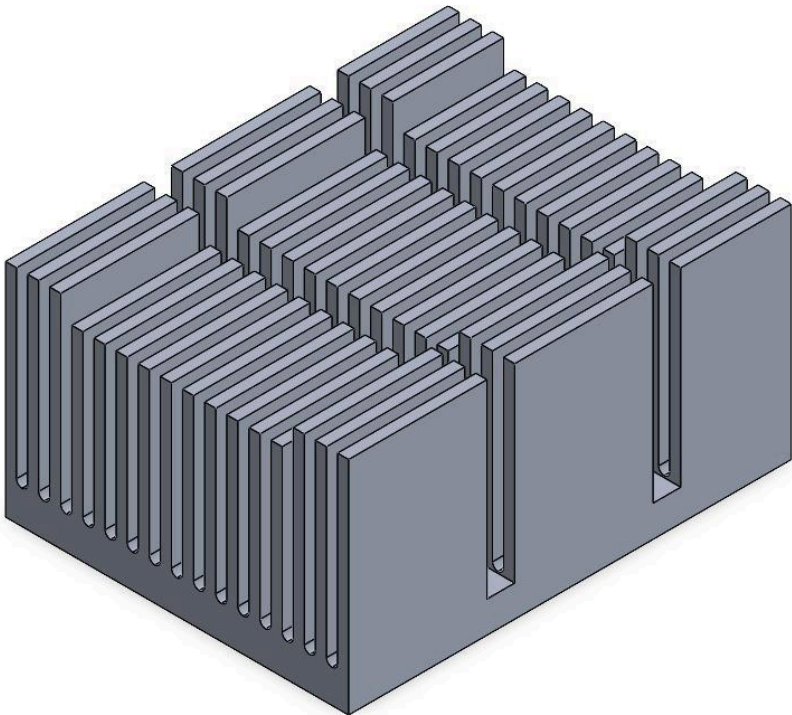
Model Information

Component Function and Role: Multi-fin heatsink designed to extract and dissipate thermal power (Q) from microprocessors. It includes a solid conduction baseplate, vertical cooling fin arrays to maximize total surface area, and a centrally split 0.012 m x 0.012 m contact face.

- Volumetric and Mass Properties:
 - Volume: 0.000100567 m³
 - Mass: 0.271531 kg (0.599 lb)
 - Density (ρ): 2,700 kg/m³
 - Weight: 2.661 N
- Material Properties (1060 Aluminum Alloy):
 - Thermal Conductivity (k): 200 W/(m-K)
 - Specific Heat (C_p): 900 J/(kg-K)
 - Mass Density (ρ): 2,700 kg/m³
 - Elastic Modulus (E): 6.90e+10 N/m²
 - Poisson's Ratio (ν): 0.33
 - Thermal Expansion Coefficient: 2.40e-05 /K
- Thermal Loads and Boundaries:
 - Heat Power-1: $Q = 50$ W applied to the 0.012 m x 0.012 m bottom split face (1 face).
 - Convection-1: Convection coefficient $h = 25$ W/(m²-K) applied across all 239 exposed faces with an ambient bulk temperature of $T_{\text{amb}} = 300$ K (27 C).
- Mesh and Discretization:
 - Mesh Type: Solid tetrahedral high-quality mesh using the Standard Mesher with 16 Jacobian points.
 - Global Element Size: 0.0029571 m (2.96 mm) with a tolerance of 0.000147855 m.
 - Element Count: 89,744 nodes and 49,101 elements.
 - Mesh Quality: 87.8% of elements have an aspect ratio under 3, with 0% distorted elements.



Heatsink



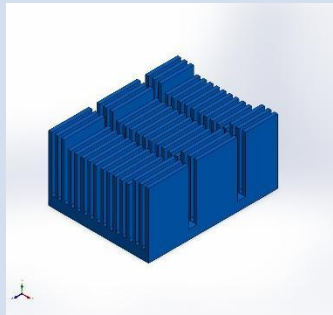
Model name: Heatsink
Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
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Heatsink

Split Line2 	Solid Body	Mass:0.271531 kg Volume:0.000100567 m³ Density:2,700 kg/m³ Weight:2.661 N	Dec 10 18:07:17 2025
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Study Properties

Study name	HEATSINK STEADY STATE ANALYSIS
Analysis type	Thermal(Steady state)
Mesh type	Solid Mesh
Solver type	FFEPlus
Solution type	Steady state
Contact resistance defined?	No
Result folder	SOLIDWORKS document

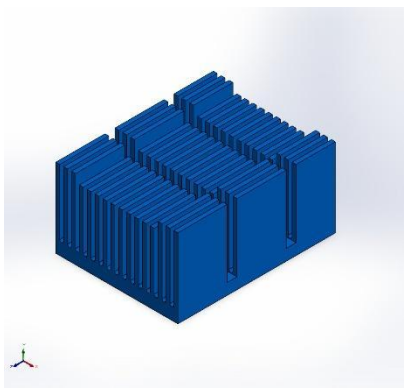
Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

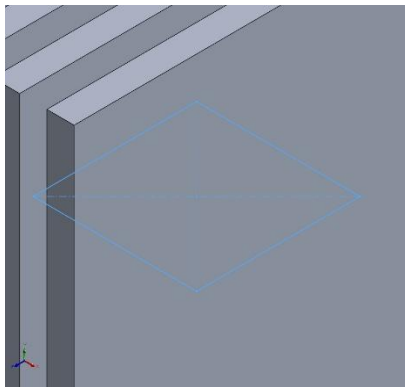


Heatsink

Material Properties

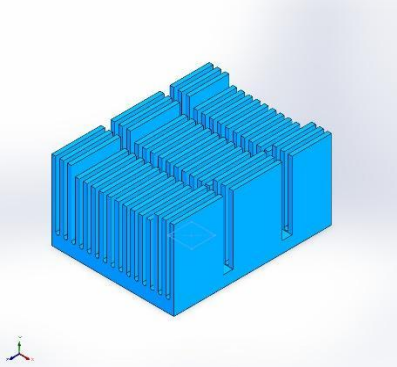
Model Reference	Properties	Components
	Name: 1060 Alloy Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Thermal conductivity: 200 W/(m.K) Specific heat: 900 J/(kg.K) Mass density: 2,700 kg/m ³	SolidBody 1(Split Line2)(Heatsink)
Curve Data:N/A		

Thermal Loads

Load name	Load Image	Load Details
Heat Power-1		Entities: 1 face(s) Heat Power Value: 50 W



Heatsink

Convection-1		Entities: 239 face(s) Convection 25 Coefficient: W/(m^2.K) Time variation: Off Temperature Off variation: Bulk Ambient 300 Kelvin Temperature: Time variation: Off
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Interaction Information

- **Component Architecture:** The heatsink is modeled and analyzed as a single, continuous solid body rather than an assembly of joined components.
- **Contact Sets:** Because there are no internal mating surfaces, bonded joints, or sliding parts, no contact sets or interaction formulations were defined.
- **Report Designation ("No Data"):** The software displays "No Data" because thermal contact formulations only apply to multi-body assemblies. Heat moves continuously through the solid material without interfacial resistance.

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Standard mesh
Automatic Transition:	Off
Include Mesh Auto Loops:	Off



Heatsink

Jacobian points for High quality mesh	16 Points
Element Size	0.0029571 m
Tolerance	0.000147855 m
Mesh Quality	High

Mesh information - Details

Total Nodes	89744
Total Elements	49101
Maximum Aspect Ratio	10.514
% of elements with Aspect Ratio < 3	87.8
Percentage of elements with Aspect Ratio > 10	0.00407
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:11
Computer name:	ANT

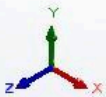
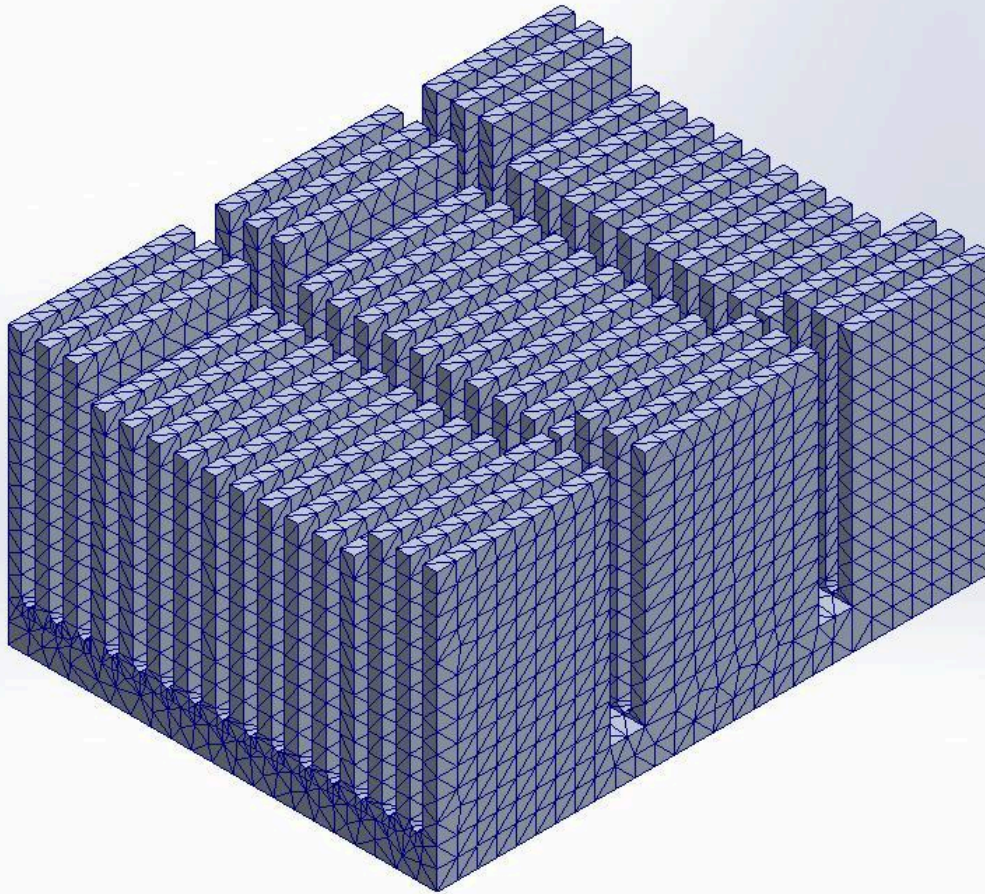
Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



Heatsink

Model name: Heatsink
Study name: HEATSINK STEADY STATE ANALYSIS(-Default-)
Plot type: Mesh Quality1



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Heatsink-HEATSINK STEADY STATE ANALYSIS-Quality-Quality1



SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of Heatsink

Heatsink

Sensor Details

- Sensor Scope: No discrete numerical tracking sensors (such as local history probes or maximum temperature flag sensors) were activated for this study.
- Report Designation ("No Data"): The simulation software automatically displays "No Data" because the full temperature field was solved across the entire body in one steady-state step using the FFEPlus solver. Maximum and minimum values are tracked directly across all nodes via the contour plots.

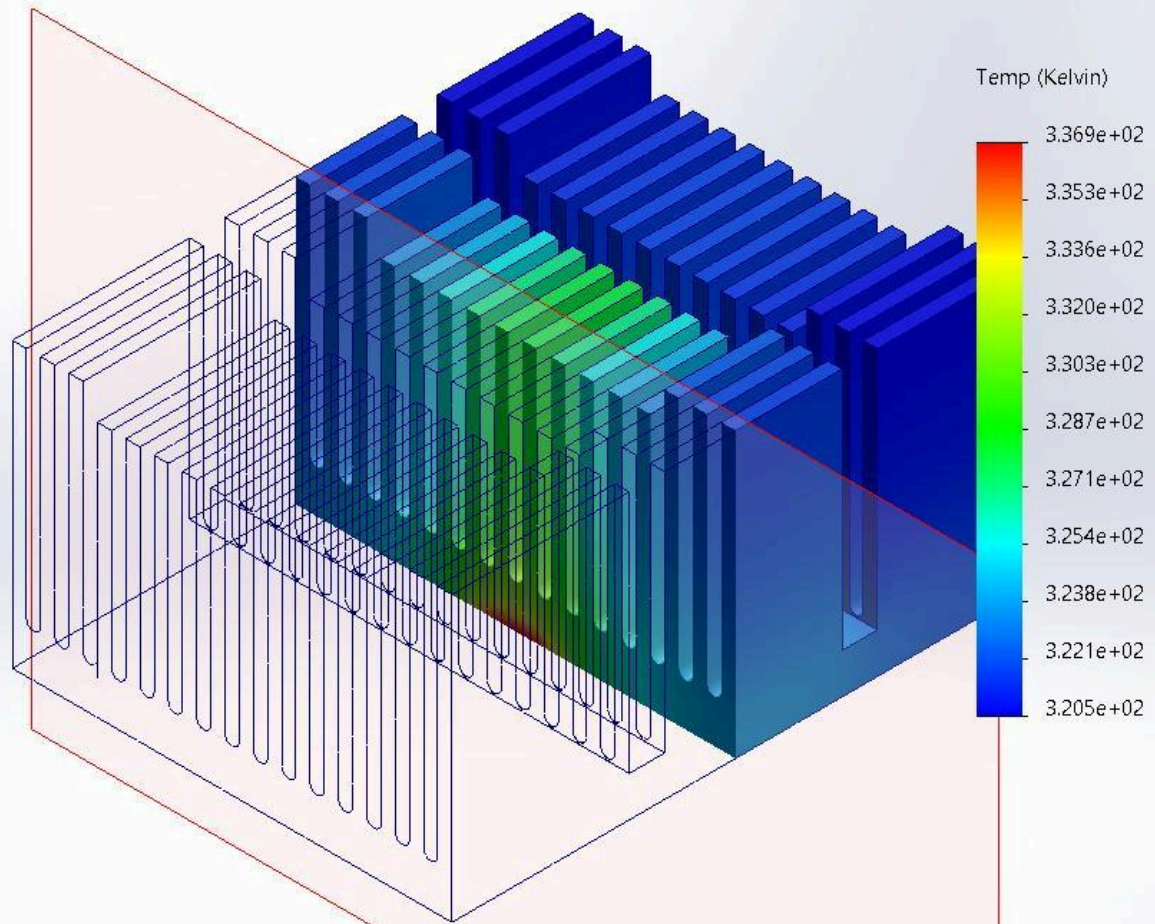
Study Results

Name	Type	Min	Max
Thermal1	TEMP: Temperature	3.205e+02Kelvin Node: 3705	3.369e+02Kelvin Node: 7520



Heatsink

Model name: Heatsink
Study name: HEATSINK STEADY STATE ANALYSIS(-Default-)
Plot type: Thermal Thermal1
Time step: 1



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Heatsink-HEATSINK STEADY STATE ANALYSIS-Thermal-Thermal1



SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of Heatsink

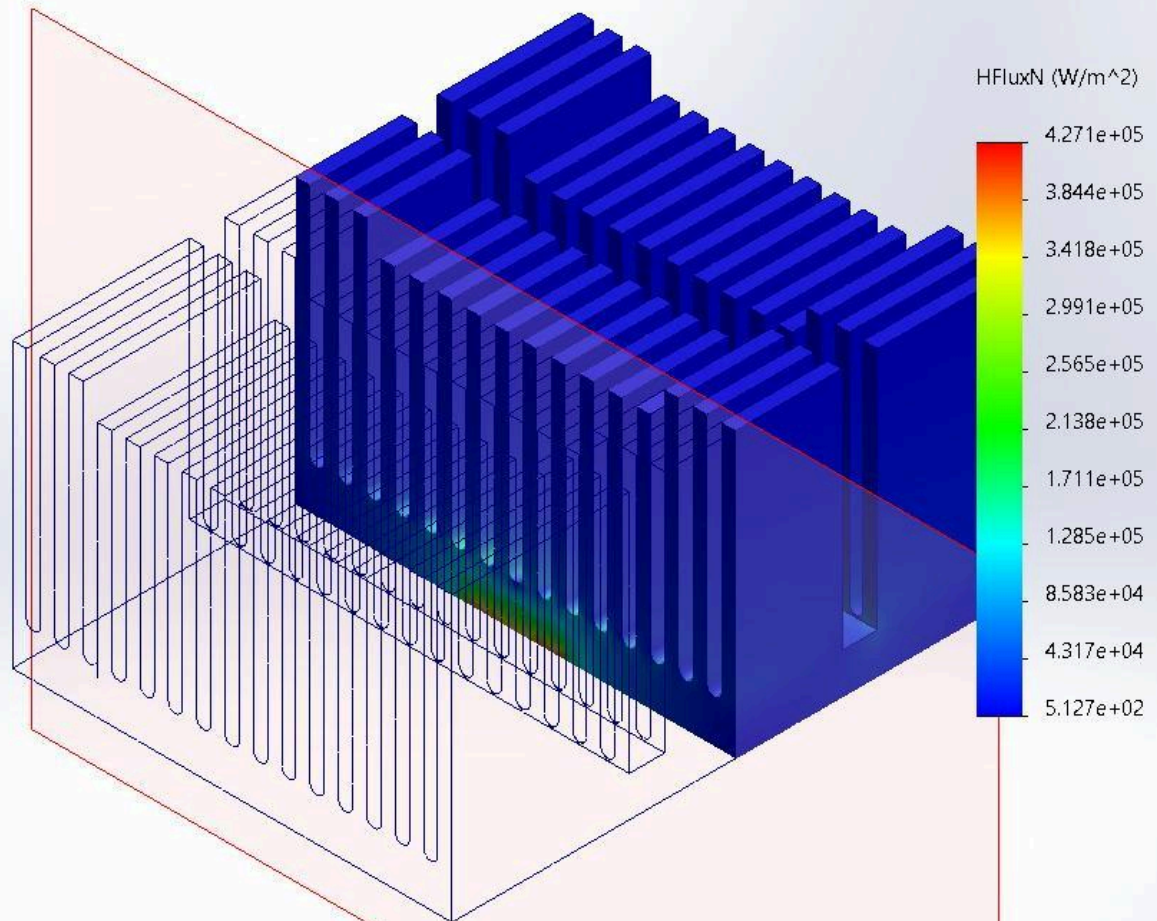
Heatsink

Name	Type	Min	Max
Thermal2	HFLUXN: Resultant heat flux	5.127e+02W/m^2 Node: 67087	4.271e+05W/m^2 Node: 89074



Heatsink

Model name: Heatsink
Study name: HEATSINK STEADY STATE ANALYSIS(-Default-)
Plot type: Thermal Thermal2
Time step: 1



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Heatsink-HEATSINK STEADY STATE ANALYSIS-Thermal-Thermal2



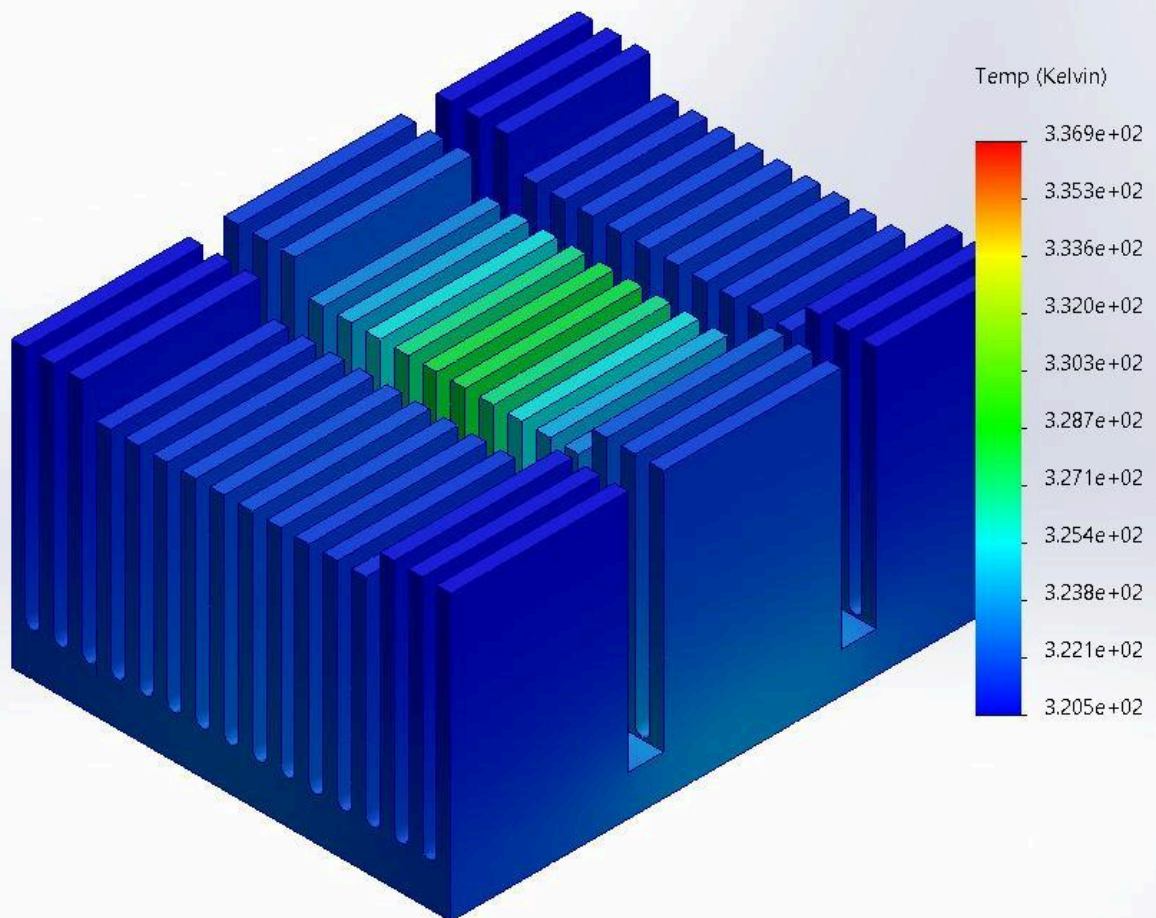
SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of Heatsink

Heatsink

Model name: Heatsink
Study name: HEATSINK STEADY STATE ANALYSIS(-Default-)
Plot type: Thermal Thermal1
Time step: 1



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Image-2



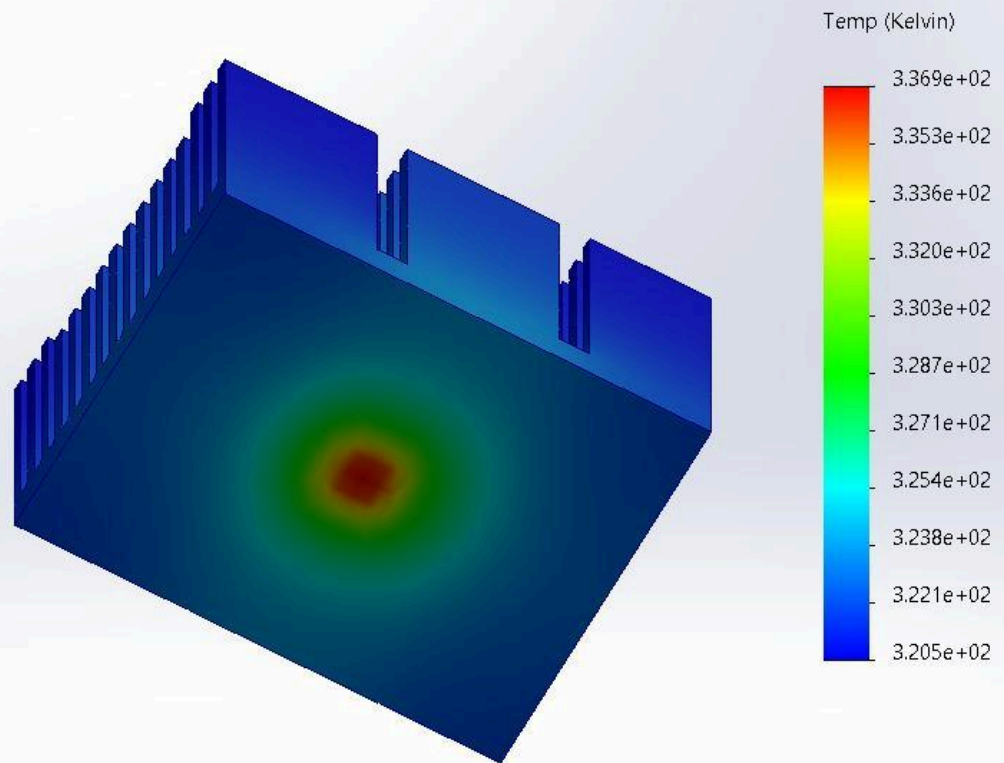
SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of Heatsink

Heatsink

Model name: Heatsink
Study name: HEATSINK STEADY STATE ANALYSIS(-Default-)
Plot type: Thermal Thermal1
Time step: 1



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Image-3



SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of Heatsink

Heatsink

Conclusion

- Temperature Results:
 - Under a steady 50 W heat load and natural air convection ($h = 25 \text{ W}/(\text{m}^2\text{-K})$), the maximum steady-state temperature reached 336.9 K (63.9 C) at Node 7520, located in the center of the CPU contact patch.
 - The minimum temperature recorded was 320.5 K (47.5 C) at Node 3705 on the outer tips of the corner cooling fins.
 - The total temperature drop across the entire heatsink was only $\Delta T = 16.4 \text{ K}$ (16.4 C), confirming that the high thermal conductivity of the 1060 aluminum alloy spreads heat rapidly across the base.
- Heat Flux Results:
 - Resultant heat flux peaked at $4.271\text{e}+05 \text{ W}/\text{m}^2$ (427,100 W/m^2) at Node 89074, located along the perimeter boundary of the CPU square where heat turns outward into the baseplate.
 - The minimum heat flux dropped to $5.127\text{e}+02 \text{ W}/\text{m}^2$ (512.7 W/m^2) at Node 67087 near the outer fin edges.
 - Section views confirm heat flows radially outward through the baseplate before traveling upward into the fin array.
- Engineering Assessment:
 - The maximum operating temperature of 63.9 C stays well below typical silicon throttling thresholds (85 C to 100 C), verifying that this heatsink design operates safely under steady-state natural convection.
 - The central fin channels operate warmer (330 K to 334 K) than the perimeter fins (320 K to 325 K). For forced convection setups (such as adding a cooling fan with $h = 100 \text{ W}/(\text{m}^2\text{-K})$), directing airflow straight down the center channels will yield the highest cooling performance.

